

SSH Logs Security Monitoring Dashboard Using Splunk

Abstract

The SSH Logs Security Monitoring Dashboard Using Splunk is developed to monitor and analyze SSH authentication logs in real time. SSH is commonly used for secure remote access, but it is also a major target for brute force attacks and unauthorized login attempts.

This project uses Splunk Enterprise to collect, process, and visualize SSH log data. The dashboard displays total SSH events, successful logins, failed login attempts, and invalid user attempts. It also detects possible brute force attacks by identifying repeated failed authentication attempts from specific IP addresses.

Geo-location mapping is implemented to identify the country origin of suspicious login attempts. The dashboard improves security monitoring, enhances threat detection, and provides better visibility into SSH activities.

Objectives

- Monitor SSH authentication activity
- Detect brute force attacks
- Identify suspicious IP addresses
- Analyze login trends over time
- Visualize attacker geo-location
- Provide centralized security dashboard

Tools & Technologies Used

- Platform: Splunk Enterprise
- Log Format: JSON SSH logs
- Server Environment: LinuxServer
- Visualization Types:
 - Single Value Panels
 - Bar Charts
 - Statistics Tables
 - Choropleth Map
- SPL (Search Processing Language)

Introduction

The **SSH Logs Dashboard** is a security monitoring project developed using Splunk Enterprise.

This dashboard analyzes SSH authentication logs to monitor:

- Total SSH activity

- Successful login attempts
- Failed login attempts
- Invalid user attempts
- Possible brute force attacks
- Geo-location of attackers

The goal of this project is to detect suspicious login behavior and improve server security monitoring.

Lab Setup

- Logs Source: ssh_logs.json
- Host: LinuxServer
- Sourcetype: _json
- Shared Time Picker Token: time_range

All panels use a common **Time Range filter** for consistency.

Dashboard Panels Description

. Authentication Overview

1. Total SSH Events

Displays total SSH log events in selected time range.

```
source="ssh_logs.json" host="LinuxServer" sourcetype="_json"
```

```
| stats count AS "Total SSH Events"
```

2. Successful Logins

Displays total successful SSH login attempts.

```
source="ssh_logs.json" host="LinuxServer" sourcetype="_json" event_type="Successful SSH Login"
```

```
| stats count AS "Successful Logins"
```

3. Failed Logins

Displays total failed login attempts.

```
source="ssh_logs.json" host="LinuxServer" sourcetype="_json" event_type="Failed SSH Login"
```

```
| stats count AS "Failed Login"
```

4. Invalid User Attempts

Displays login attempts using invalid usernames.

```
index=auth "sshd" "invalid user"
| stats count AS "Invalid User Attempts"
```

Login Activity Trends

1. Failed Logins by Username (Bar Chart)

This panel shows which usernames are most targeted.

```
source="ssh_logs_new.json" host="LinuxNew" sourcetype="_json" event_type="Failed SSH Login"
| top username
```

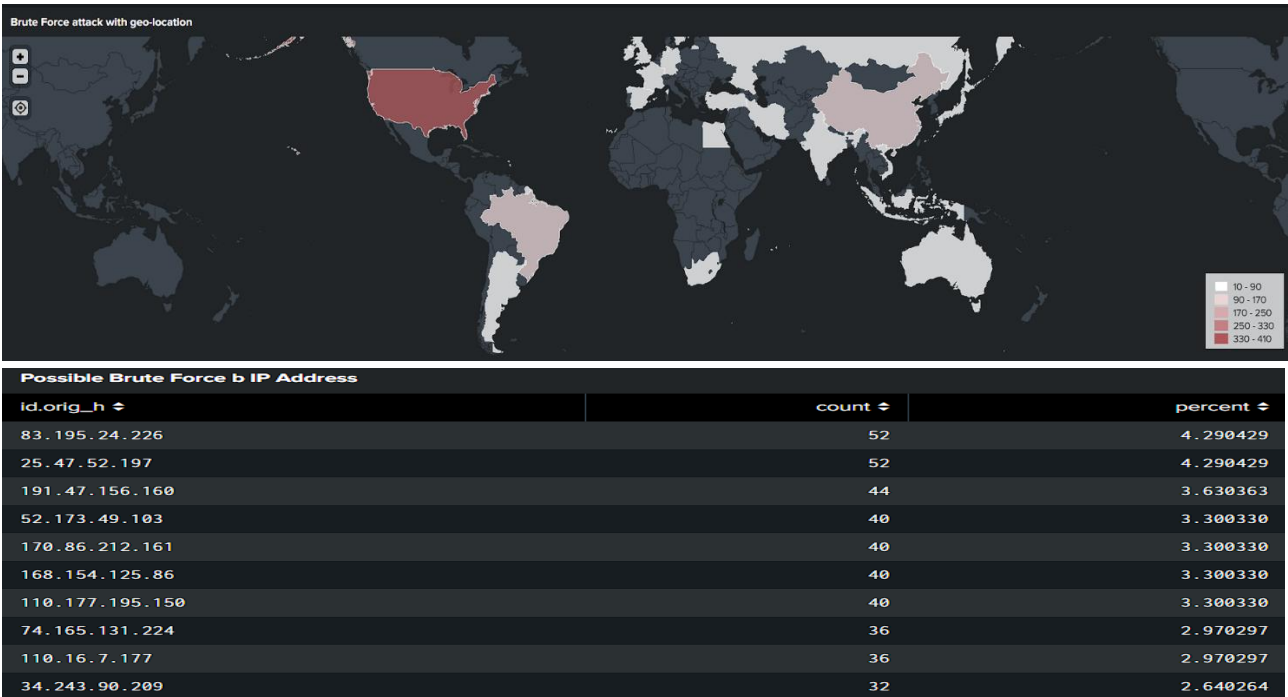
2. Possible Brute Force by IP (Statistics Table)

This panel identifies IP addresses with multiple failed attempts.

```
source="ssh_logs_new.json" host="LinuxNew" sourcetype="_json" event_type="Multiple Failed Authentication Attempts"
| top id.orig_h
```

High count from a single IP may indicate a brute force attack.

3. Brute Force Attack Geo-Location



This panel visualizes attacker IP addresses by country using geo-location mapping.

```
source="ssh_logs_new.json" host="LinuxNew" sourcetype="_json" event_type="Multiple Failed Authentication Attempts"
| table id.orig_h
| iplocation id.orig_h
```

| stats count by Country

| geom geo_countries featureIdField="Country"

- iplocation converts IP to country
- geom displays the data on world map

Darker color indicates higher attack frequency.

Key Features

- Centralized SSH monitoring
- Real-time attack detection
- Brute force identification
- Geographic attack visualization
- Interactive time filtering

Results & Findings

- Multiple failed login attempts detected from specific IP addresses
- Certain usernames were targeted frequently
- Geo-location analysis showed attack sources from multiple countries
- Possible brute force patterns were identified

Conclusion

The **SSH Logs Dashboard** successfully monitors and analyzes SSH authentication activity using Splunk Enterprise. It helps security teams detect brute force attacks, monitor login trends, and identify attacker locations. This dashboard improves overall system security visibility and proactive threat detection.

